

Կոնֆիգուրացիա VS 2026 + x64 համար cmake -G "Visual Studio 18 2026" -A x64 -S .. -B .#
Build անել Debug կոնֆիգուրացիայովcmake --build . --config Debug

```
x64 Native Tools Command P...
C:\Users\user\Desktop\Software Security\Lab3>cd "C:\Users\user\Desktop\Software Security\Lab3\build_vs"

C:\Users\user\Desktop\Software Security\Lab3\build_vs>cmake -G "Visual Studio 18 2026" -A x64 -S .. -B .
-- Selecting Windows SDK version 10.0.26100.0 to target Windows 10.0.26200.
-- The C compiler identification is MSVC 19.50.35727.0
-- The CXX compiler identification is MSVC 19.50.35727.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: C:/Program Files/Microsoft Visual Studio/18/Community/VC/Tools/MSVC/14.50.35717/bin/Hostx64/x64/cl.exe - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: C:/Program Files/Microsoft Visual Studio/18/Community/VC/Tools/MSVC/14.50.35717/bin/Hostx64/x64/cl.exe - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (6.9s)
-- Generating done (0.8s)
-- Build files have been written to: C:/Users/user/Desktop/Software Security/Lab3/build_vs

C:\Users\user\Desktop\Software Security\Lab3\build_vs>cmake --build . --config Debug
Версия MSBuild 18.4.0+6e61e96ac для .NET Framework

1>Checking Build System
Building Custom Rule C:/Users/user/Desktop/Software Security/Lab3/CMakeLists.txt
main.cpp
HardwareID.cpp
C:\Users\user\Desktop\Software Security\Lab3\src\HardwareID.cpp(11,9): warning C4805: WIN32_LEAN_AND_MEAN: изменение ма
кроопределения [C:\Users\user\Desktop\Software Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\HardwareID.cpp(11,9):
"WIN32_LEAN_AND_MEAN" ранее объявлен в командной строке

LicenseKey.cpp
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseKey.cpp(34,34): warning C4244: :: преобразование "time_t" в "un
signed long": возможна потеря данных [C:\Users\user\Desktop\Software Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseKey.cpp(68,36): warning C4996: 'ctime': This function or variab
le may be unsafe. Consider using ctime_s instead. To disable deprecation, use _CRT_SECURE_NO_WARNINGS. See online help
for details. [C:\Users\user\Desktop\Software Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseKey.cpp(71,41): warning C4996: 'ctime': This function or variab
le may be unsafe. Consider using ctime_s instead. To disable deprecation, use _CRT_SECURE_NO_WARNINGS. See online help

C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(87,22): warning C4566: символ, представленный униве
рсальным именем символа "u056c", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u0532", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u0548", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u053c", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u0559", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u053b", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u0551", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u0535", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u0546", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u0536", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u0531", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
C:\Users\user\Desktop\Software Security\Lab3\src\LicenseManager.cpp(91,18): warning C4566: символ, представленный униве
рсальным именем символа "u0538", невозможно представить в текущей кодовой странице (1251) [C:\Users\user\Desktop\Softw
are Security\Lab3\build_vs\LicensingSystem.vcxproj]
Создание кода...
LicensingSystem.vcxproj -> C:\Users\user\Desktop\Software Security\Lab3\build_vs\Debug\LicensingSystem.exe
Building Custom Rule C:/Users/user/Desktop/Software Security/Lab3/CMakeLists.txt

C:\Users\user\Desktop\Software Security\Lab3\build_vs>
```

Կիրառում ենք մեկույուսում արդեն առկա գործիքները

```
x64 Native Tools Command P...
C:\Users\user\Desktop\Software Security\Lab3\build_vs\Debug>.\\LicensingSystem.exe
=====
LICENSE SYSTEM - LABORATORY WORK
=====

Current Device ID 0000000076F677B8
CPU ID 49656E6900000020BFCBDF00096A4
MAC address 00:15:5d:01:0f:00

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
0. Exit
Your choice: 1
Enter licensee name: Ss
=====
!!!! PERPETUAL LICENSE CREATED:
=====
License Key: 40F3F-5BD40-F3F50
Licensee: Ss
Issue Date: Wed Mar 18 04:37:48 2026
Type: PERPETUAL
=====

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
0. Exit
Your choice: 4
????? ??????????????
=====
License Key: 40F3F-5BD40-F3F50
Licensee: Ss
Issue Date: Wed Mar 18 04:37:48 2026
Type: PERPETUAL
=====

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
Your choice: 3
Enter license key: 40F3F-5BD40-F3F50
v LICENSE IS VALID:

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
0. Exit
Your choice: 2
Enter licensee name: AMrr
=====
v TEMPORARY LICENSE CREATED (30 days):
=====
License Key: 7947A-3F874-47A3F
Licensee: AMrr
Issue Date: Wed Mar 18 04:38:40 2026
Valid until: Fri Apr 17 04:38:40 2026
Type: TEMPORARY
=====

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
0. Exit
Your choice: 4
????? ??????????????
=====
License Key: 40F3F-5BD40-F3F50
Licensee: Ss
Issue Date: Wed Mar 18 04:37:48 2026
Type: PERPETUAL
=====
License Key: 7947A-3F874-47A3F
Licensee: AMrr
Issue Date: Wed Mar 18 04:38:40 2026
Valid until: Fri Apr 17 04:38:40 2026
Type: TEMPORARY
=====

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
```

Ավելացնում ենք 6-րդ կետը, որպեսզի հնարավորություն ունենանք փոփոխել ժամանակավոր

բանալիների տևողության օրերը

main.cpp

```
#include <iostream>#include <string>#include "LicenseManager.h"int main() {
```

```

std::cout << "===== " << std::endl;
std::cout << " LICENSE SYSTEM - LABORATORY WORK " << std::endl;
std::cout << "===== " << std::endl;
std::cout << std::endl;

LicenseManager manager;

int choice;

std::string userName;

int tempLicenseDays = 30; // Default 30 days for temporary licensesdo {

std::cout << "\n--- SELECT ACTION ---

" << std::endl;

std::cout << "1. Create perpetual license" << std::endl;

std::cout << "2. Create temporary license (" << tempLicenseDays << " days)" << std::endl;

std::cout << "3. Verify license" << std::endl;

std::cout << "4. Show all licenses" << std::endl;

std::cout << "5. Save license to file" << std::endl;

std::cout << "6. Change temporary license duration (current: " << tempLicenseDays << " days)"
<< std::endl;

std::cout << "0. Exit" << std::endl;

std::cout << "Your choice: ";

std::cin >> choice;

std::cin.ignore();

switch (choice) {

case 1: {

std::cout << "Enter licensee name: ";

std::getline(std::cin, userName);

```

```
LicenseKey license = manager.createPerpetualLicense(userName);

std::cout << "\n!!!! PERPETUAL LICENSE CREATED:" << std::endl;

license.printInfo();

break;

}

case 2: {

std::cout << "Enter licensee name: ";

std::getline(std::cin, userName);

LicenseKey license = manager.createTemporaryLicense(userName, tempLicenseDays);

std::cout << "\n!!!! TEMPORARY LICENSE CREATED (" << tempLicenseDays << " days)." <<
std::endl;

license.printInfo();

break;

}

case 3: {

std::string key;

std::cout << "Enter license key: ";

std::getline(std::cin, key);

if (manager.verifyLicense(key)) {

std::cout << "\v LICENSE IS VALID" << std::endl;

}

else {

std::cout << "\x LICENSE IS INVALID OR NOT FOUND" << std::endl;

}

break; }
```

```

case 4: {
manager.listAllLicenses();

break; }case 5: {

std::string key, filename;

std::cout << "Enter license key: ";

std::getline(std::cin, key);

std::cout << "Enter filename: ";

std::getline(std::cin, filename);

LicenseKey temp(manager.getCurrentMachineID(),

"Test User");

if (manager.saveToFile(temp, filename)) {

std::cout << "v License saved to " << filename << std::endl;}

else {

std::cout << "x Error saving to file" << std::endl;

}

break; }

case 6: {

std::cout << "Enter new duration in days for temporary license: ";

int newDays;

std::cin >> newDays;

std::cin.ignore();

if (newDays > 0) {

tempLicenseDays = newDays;

std::cout << "Temporary license duration updated to " << tempLicenseDays << " days.

```

```
" << std::endl;

}

else {

std::cout << "Invalid number of days.

" << std::endl;}

break;}

case 0: {

std::cout << "Program finished.

" << std::endl;

break;}

default: {

std::cout << "Invalid choice" << std::endl;

break;}

}

} while (choice != 0);

return 0;}
```

```

Type: TEMPORARY
=====
--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (30 days)
3. Verify license
4. Show all licenses
5. Save license to file
6. Change temporary license duration (current: 30 days)
0. Exit
Your choice: 6
Enter new duration in days for temporary license: 6
Temporary license duration updated to 6 days.

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (6 days)
3. Verify license
4. Show all licenses
5. Save license to file
6. Change temporary license duration (current: 6 days)
0. Exit
Your choice: 4

????? ??????????:
=====
License Key: 05260-68205-26068
Licensee: VSS
Issue Date: Wed Mar 18 11:52:12 2026
Valid until: Fri Apr 17 11:52:12 2026
Type: TEMPORARY
=====

--- SELECT ACTION ---
1. Create perpetual license
2. Create temporary license (6 days)
3. Verify license
4. Show all licenses
5. Save license to file
6. Change temporary license duration (current: 6 days)

```